Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (ODD)**	**Course Name**	**Web Technology Lab**
Semester	**3**	**Course Code**	**23CT1301**
Roll No	**13**	**Name of Student**	**Juhi Parwani**

Practical Number	1(A)
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	To explore Git and GitHub for version control and learn how to manage source code through commits, branches, merging, remote repositories, pull requests, and conflict handling.
Problem Definition	To understand how Git maintains a history of project changes and how GitHub supports online code storage and teamwork. The practical covers repository setup, file staging, committing updates, creating and merging branches, transferring changes between local and remote repositories, cloning projects, and working with pull requests. It also demonstrates how to identify and handle conflicts when changes overlap in the same file.
Theory (100 words)	Git is a tool that records changes made to files throughout a project's development. It helps developers review previous versions, organize their work into branches, and combine updates when required. GitHub provides an online location for Git repositories, making it easier to share code and review contributions. Branches allow developers to work on separate features without directly changing the main branch. Commits save snapshots of changes, while push and pull commands exchange updates with a remote repository. When two branches contain incompatible edits to the same section, Git

	may report a merge conflict that needs to be resolved before the changes can be combined.
Procedure and Execution (100 Words)	Step for Implementation: 1. Install Git and set the required user details. 2. Open the project directory and initialize the repository. 3. Add the project files to staging and save them using a commit. 4. Create a remote repository on GitHub. 5. Link the local repository with GitHub and upload the project. 6. Create a feature branch and make changes to project files. 7. Open a pull request and merge the approved changes. 8. Clone the repository into another folder to practise working with a copy. 9. Edit the same file differently on two branches to produce a merge conflict. 10. Resolve the conflict, commit the correction, and push the updated branch. The practical project was organized in the `Practical_9` directory with these files: • `index.html` • `style.css` • `script.js` • `README.md` • `branch-demo.txt` • `conflict-demo.txt`
	Code: # 1. Check Git version git --version # 2. Configure username and email git config --global user.name "Juhi Parwani" git config --global user.email "juhiparwani65@gmail.com" # 3. Initialize and check repository git init git status

```
# 4. Stage and commit files
git add .
git commit -m "Initial commit"
git log --oneline

# 5. Create and switch branches
git branch
git switch -c practical9-feature

# 6. Push changes to GitHub
git push -u origin practical9-feature

# 7. Pull changes from GitHub
git pull origin main

# 8. Clone repository
git clone https://github.com/juhiparwa-1507/Web-Technology-Lab.git

# 9. Merge branches
git switch main
git merge practical9-feature

# 10. Check repository status
git status

# Switch to the feature branch
git switch conflict-student-a

# Merge the other branch
git merge conflict-student-b

# After manually resolving the conflict
git add Practical_9/conflict-demo.txt
git commit -m "Resolve merge conflict"

# Push resolved changes
git push origin conflict-student-a
```

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	Output: The project repository was managed using Git, and its changes were uploaded to GitHub. The practical demonstrated file staging, commits, branch creation, branch merging, and collaboration through a pull request. A separate copy of the repository was created using the clone command. A merge conflict was also produced during branch integration, followed by a resolution commit and a push to the remote branch. The pull command indicated that the local repository was up to date with the specified remote branch. The cloned repository's status showed that its working tree was clean.
Output Analysis	The results demonstrate how Git tracks project modifications and maintains a record of saved changes. Branches make it possible to develop and test updates separately, while merging and pull requests provide a way to integrate contributions. The conflict exercise illustrates why overlapping edits sometimes require manual review before they can be combined. Cloning also makes it possible to work with a separate local copy of a GitHub repository.
Link of student Github profile where lab assignment has been uploaded	**https://github.com/juhiparwa-1507/Web-Technology-Lab.git**
Conclusion	This practical provided experience with Git-based version control and GitHub collaboration. Repository management, committing changes, branching, merging, cloning, remote synchronization, pull requests, and merge-conflict handling were practised as part of the workflow.
Plag Report (Similarity index < 12%)	

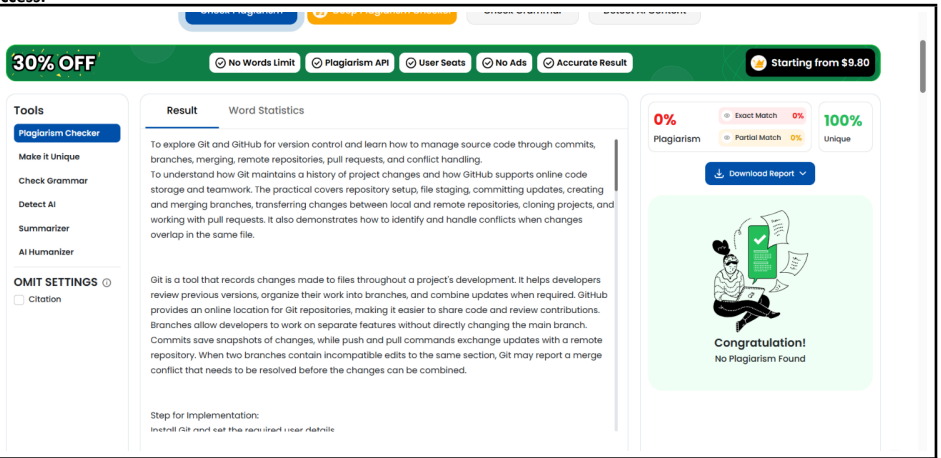
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Date	**14-10-26**